Department of Mathematics and Natural Sciences
Section - 06, Quiz - 03, Spring 2025
Course Code: MAT110, Course Title: MATH I: Differential Calculus and Co-ordinate Geometry

Time: 50 minutes | Marks: 25

N.B. Answer all of the following questions. Make sure to show your steps and calculations as appropriate.

1.	Given $f(x,y) = e^{xy}(\tan x + \sec y)$, find $L(x,y)$ and $Q(x,y)$ at $P(0,0)$.	8
2.	Find all the relative extrema and saddle points, if any, of the function $f(x,y) = (6x - x^2)(4y - y^2)$.	9
3.	Find the extrema of the function, $f(x,y,z) = xyz$, given the constraint: $x^2 + y^2 + z^2 = 1$.	8

–Best of luck–